

IIT HYDERABAD

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices and Circuits Laboratory Report

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1 Objective

- To plot and analyze the output (I_D vs V_{DS}) and transfer (I_D vs V_{GS}) characteristics of an N-channel MOSFET (2N7000) under safe operating limits.
- From the acquired data, determine the key device parameters including threshold voltage (V_T), transconductance (g_m), and saturation drain resistance (r_d).
- Finally, compute the overall amplification factor ($\mu = g_m \times r_d$) and verify the pinch-off operating boundary.

2 Components Required

- 2N7000 transistor - 1
- Regulated DC Power Supply
- Resistors (1 k Ω , 10 k Ω)
- Voltmeter
- Ammeter (Milli)
- Breadboard
- Connecting wires

3 Output (Drain) characteristics

3.1 Aim

To plot the output characteristics of an N-channel MOSFET (2N7000) by plotting I_D versus V_{DS} for different fixed values of V_{GS} , and to determine the drain resistance r_d in the saturation region.

3.2 Circuit

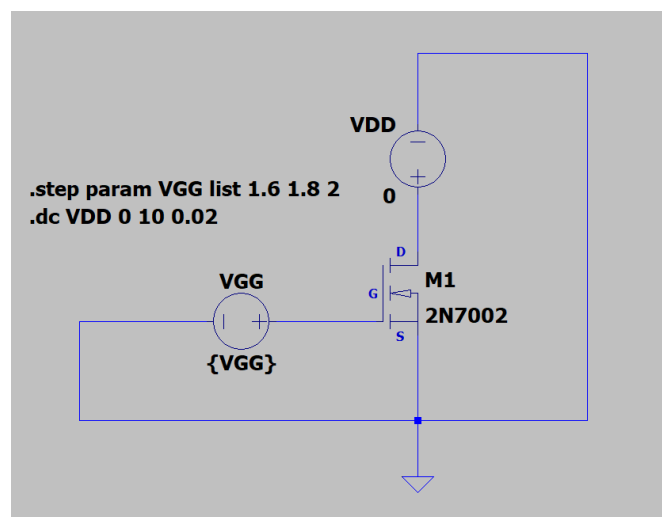


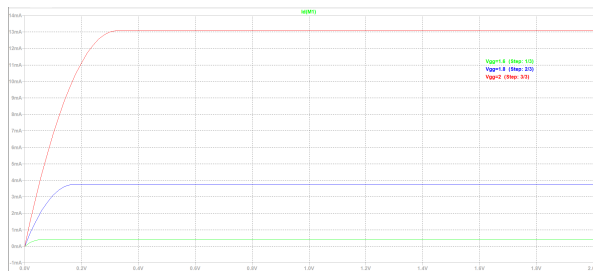
Figure 1: LTspice Circuit Schematic

3.3 Observation Table

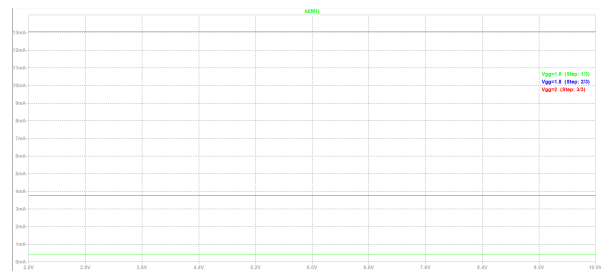
Table 1: Measured drain characteristics of the 2N7000 NMOSFET

V_{DS} (V)	I_D at $V_{GS} = 2.0$ V (mA)	I_D at $V_{GS} = 1.8$ V (mA)	I_D at $V_{GS} = 1.6$ V (mA)
0.0	0.000	0.000	0.000
0.2	7.18	1.660	0.182
0.4	8.27	1.770	0.187
0.6	8.39	1.780	0.1871
0.8	8.46	1.790	0.1872
1.0	8.49	1.799	0.1874
1.2	8.50	1.800	0.1876
1.4	8.52	1.800	0.1877
1.6	8.54	1.803	0.1884
1.8	8.56	1.804	0.1888
2.0	8.64	1.806	0.1894
3.0	8.65	1.807	0.1895
4.0	8.68	1.810	0.1900
5.0	8.74	1.820	0.1920
6.0	8.84	1.828	0.1930
7.0	8.92	1.838	0.1940
8.0	8.96	1.840	0.1943
9.0	9.09	1.846	0.1949
10.0	9.24	1.850	0.1950

3.4 Simulation plots

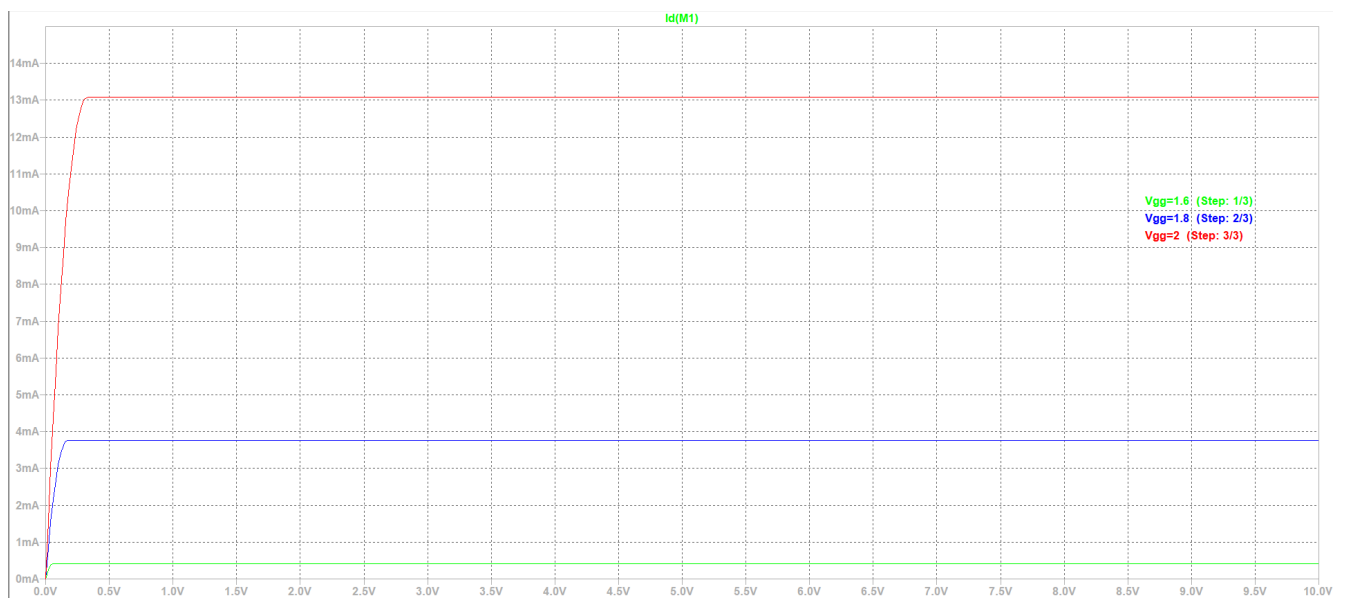


(a) low V_{DS} sweep (0-2V)



(b) high V_{DS} sweep (2-10V)

Figure 2: LTspice Simulation Curves for Output characteristics



3.5 Experimental characteristic plots

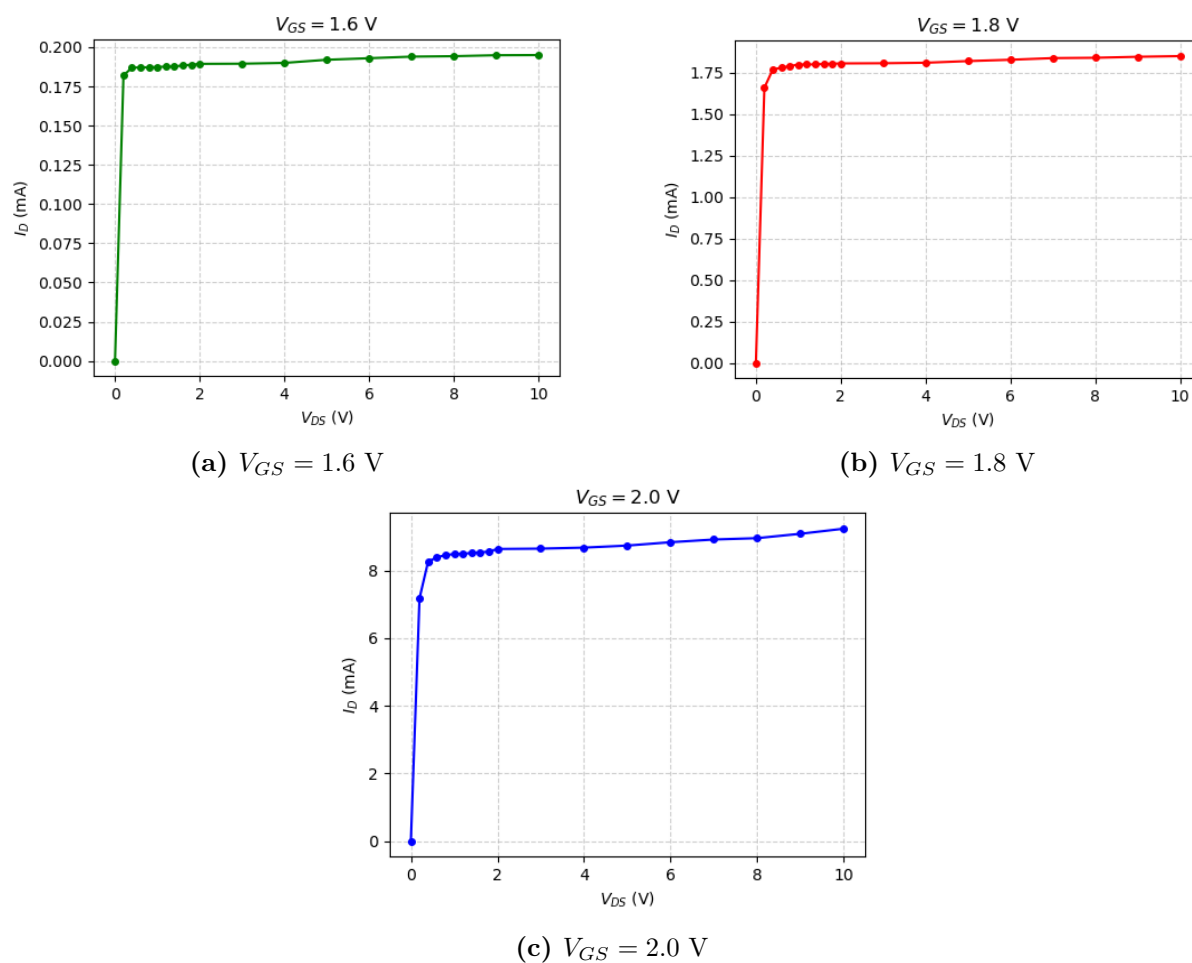
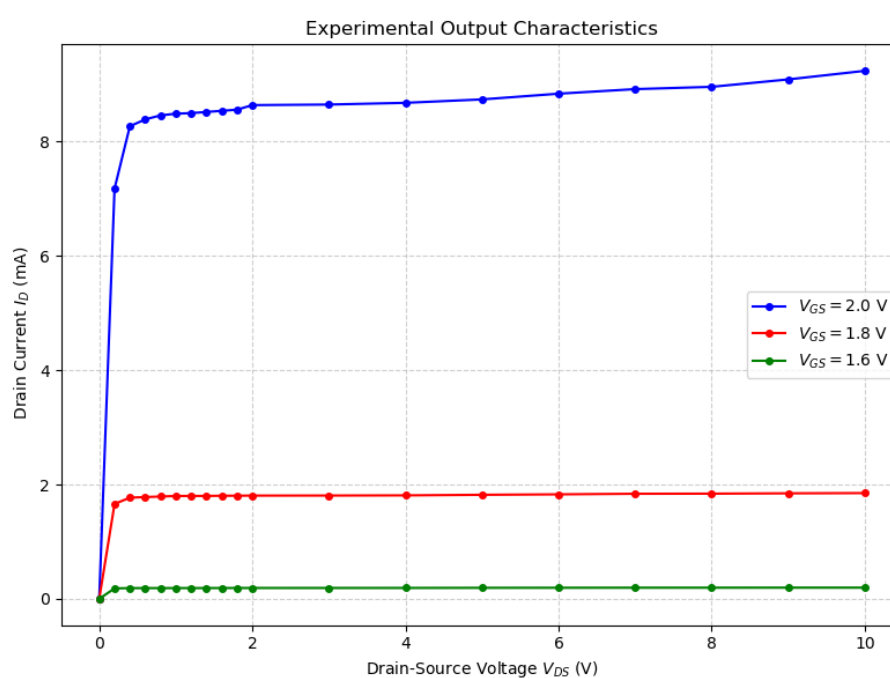


Figure 3: Output characteristics of the 2N7000 NMOSFET for different values of V_{GS} .



3.6 Calculations

3.6.1 Drain resistance

The drain resistance is calculated from the slope of the saturation region of the output characteristics:

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}$$

The points at $V_{DS} = 2\text{ V}$ and 10 V were selected as they lie well within the saturation region, beyond the knee of the output characteristics. Hence, the change in V_{DS} and I_D over this range gives a reliable estimate of r_d .

For $V_{GS} = 2.0\text{ V}$:

$$\begin{aligned} r_d &= \frac{10 - 2}{(9.24 - 8.64) \times 10^{-3}} \\ &= \frac{8}{0.60 \times 10^{-3}} \\ &= \mathbf{13.33\text{ k}\Omega} \end{aligned}$$

For $V_{GS} = 1.8\text{ V}$:

$$\begin{aligned} r_d &= \frac{10 - 2}{(1.850 - 1.806) \times 10^{-3}} \\ &= \frac{8}{0.044 \times 10^{-3}} \\ &= \mathbf{181.82\text{ k}\Omega} \end{aligned}$$

For $V_{GS} = 1.6\text{ V}$:

$$\begin{aligned} r_d &= \frac{10 - 2}{(0.1950 - 0.1894) \times 10^{-3}} \\ &= \frac{8}{0.0056 \times 10^{-3}} \\ &= \mathbf{1.43\text{ M}\Omega} \end{aligned}$$

3.6.2 Saturation Voltage $V_{DS(\text{sat})}$

The MOSFET enters the saturation region when the drain-source voltage reaches the pinch-off condition,

$$V_{DS(\text{sat})} = V_{GS} - V_T.$$

Experimentally, $V_{DS(\text{sat})}$ is determined from the output characteristics by identifying

the point at which the I_D - V_{DS} curve changes from the rapidly increasing region to the nearly flat saturation region.

From the measured output characteristics, the approximate values of $V_{DS(\text{sat})}$ were obtained as

$$V_{GS} = 2.0 \text{ V} : \quad V_{DS(\text{sat})} \approx 0.6 \text{ V},$$

$$V_{GS} = 1.8 \text{ V} : \quad V_{DS(\text{sat})} \approx 0.4 \text{ V},$$

$$V_{GS} = 1.6 \text{ V} : \quad V_{DS(\text{sat})} \approx 0.2 \text{ V}.$$

3.7 Results

Table 2: Calculated Results from Drain Characteristics

V_{GS} (V)	$V_{DS(\text{sat})}$ (V)	r_d (k Ω)	$V_{GS} - V_T$ (V)
2.0	0.6	13.33	0.6
1.8	0.4	181.82	0.4
1.6	0.2	1428.57	0.2

3.8 Key Inferences (Part-A)

- The output characteristics exhibit a linear (triode) region at low V_{DS} , followed by a saturation region where I_D varies only slightly with V_{DS} .
- The drain resistance r_d is determined from the slope of the saturation-region characteristics. A flatter curve corresponds to a higher value of r_d .
- From the measured characteristics, r_d increases as V_{GS} decreases, with the $V_{GS} = 1.6 \text{ V}$ curve showing the highest r_d .

4 Part B – Transfer (Input) Characteristics

4.1 Aim

To plot the transfer characteristics of the 2N7000 NMOSFET by plotting drain current I_D against gate-source voltage V_{GS} for fixed values of drain-source voltage V_{DS} , and to determine the threshold voltage V_T and transconductance g_m from the measured data.

4.2 Circuit

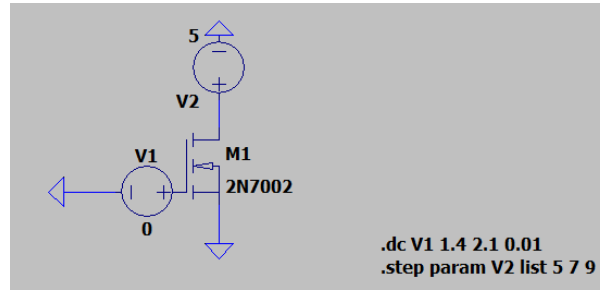


Figure 4: circuit

4.3 Observation Table

The measured values of drain current for different values of V_{GS} are shown in Table 3.

Table 3: Measured transfer characteristics of the 2N7000 NMOSFET

V_{GS} (V)	I_D at $V_{DS} = 5$ V (mA)	I_D at $V_{DS} = 7$ V (mA)	I_D at $V_{DS} = 9$ V (mA)
1.4	0.0107	0.0110	0.0110
1.5	0.0475	0.0480	0.0481
1.6	0.1940	0.1900	0.1940
1.7	0.6600	0.6650	0.6580
1.8	1.8440	1.8500	1.8500
1.9	4.4010	4.4200	4.4020
2.0	8.8000	8.8200	8.9020
2.1	15.8000	16.2000	16.4000

4.4 Experimental characteristic plot

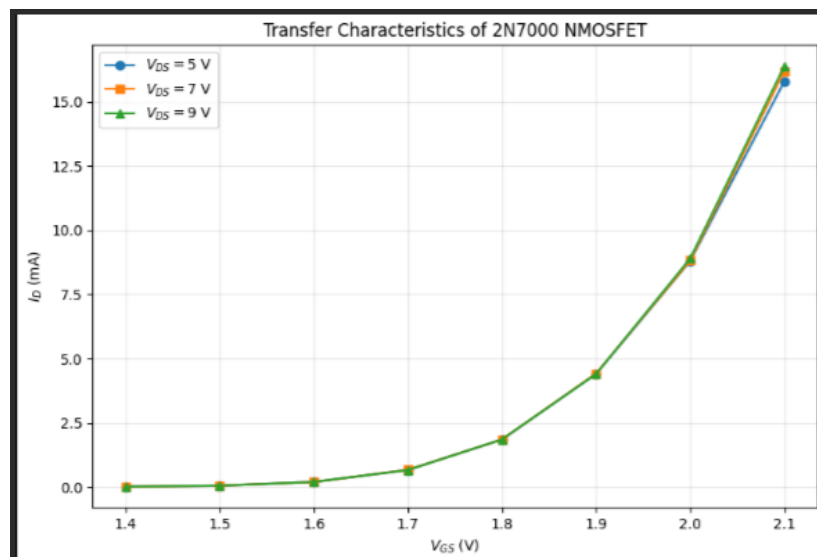


Figure 5: python plot from observations

4.5 Simulation plot

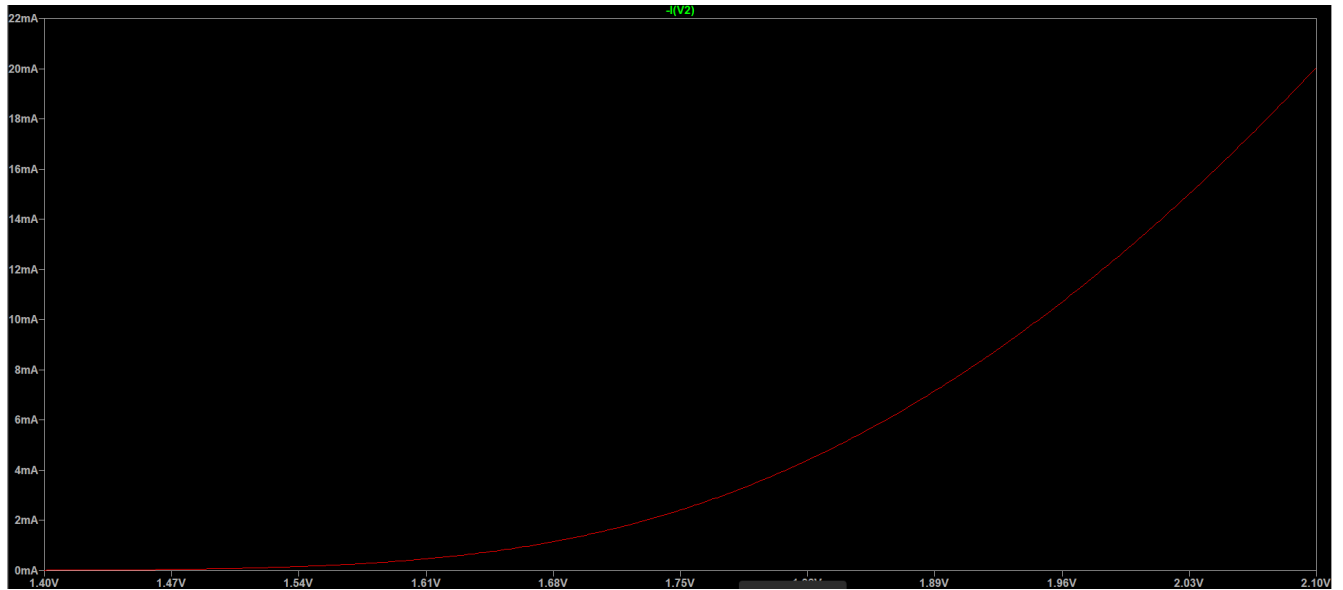


Figure 6: Simulation (The three simulated transfer characteristics nearly overlap for the values of V_{DS} .)

4.6 Theory

For an enhancement-mode NMOSFET, the drain current is approximately zero when

$$V_{GS} < V_T.$$

When the gate-source voltage exceeds the threshold voltage, an inversion channel is formed and the drain current increases rapidly.

In the saturation region, the ideal square-law relationship is

$$I_D = K(V_{GS} - V_T)^2,$$

where K is a device-dependent constant.

Taking the square root,

$$\sqrt{I_D} = \sqrt{K}(V_{GS} - V_T).$$

Thus, a plot of $\sqrt{I_D}$ versus V_{GS} should approximately be linear in the saturation region, and extrapolation of the linear portion to $\sqrt{I_D} = 0$ provides an independent estimate of the threshold voltage.

4.7 Determination of Threshold Voltage

From the measured transfer characteristics, the drain current first becomes measurably different from zero at

$$V_{GS} = 1.4 \text{ V}.$$

At this voltage, the measured drain current is approximately

$$I_D = 0.0107 \text{ mA}$$

for $V_{DS} = 5 \text{ V}$.

Therefore, based on the onset of measurable drain current,

$$V_T \approx 1.4 \text{ V}$$

This value represents the experimental threshold voltage determined directly from the transfer characteristic.

4.8 Independent Determination of V_T Using $\sqrt{I_D}$

For the saturation region,

$$I_D = K(V_{GS} - V_T)^2.$$

Therefore,

$$\sqrt{I_D} = \sqrt{K}V_{GS} - \sqrt{K}V_T.$$

Using the measured data, $\sqrt{I_D}$ was calculated and plotted against V_{GS} . The approximately linear portion was extrapolated to the V_{GS} axis.

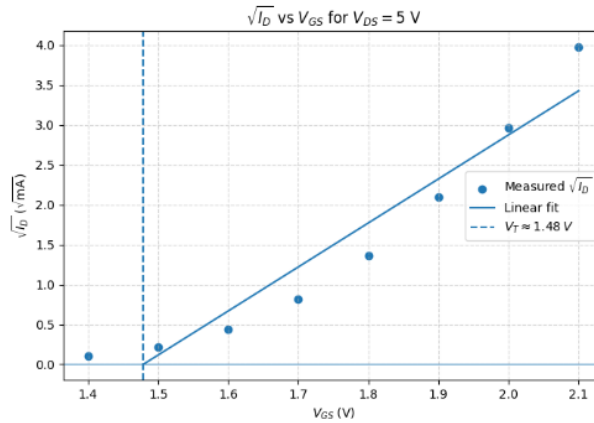


Figure 7: Plot of $\sqrt{I_D}$ versus V_{GS} for $V_{DS} = 5 \text{ V}$ for independent determination of threshold voltage

The linear fit to the measured data gives an intercept of approximately

$$V_T \approx 1.48 \text{ V}.$$

Hence, the independent estimate of the threshold voltage is

$$\boxed{V_T \approx 1.48 \text{ V}}$$

which is reasonably close to the value obtained directly from the onset of measurable current.

4.9 Determination of Transconductance

The transconductance is defined as

$$g_m = \left(\frac{\Delta I_D}{\Delta V_{GS}} \right)_{V_{DS}=\text{constant}}.$$

The transconductance was calculated at the operating point around

$$V_{GS} = 2.0 \text{ V}, \quad V_{DS} = 5 \text{ V}.$$

Two points on either side of $V_{GS} = 2.0 \text{ V}$ were selected:

$$(V_{GS1}, I_{D1}) = (1.9 \text{ V}, 4.401 \text{ mA})$$

and

$$(V_{GS2}, I_{D2}) = (2.1 \text{ V}, 15.8 \text{ mA}).$$

Therefore,

$$g_m = \frac{I_{D2} - I_{D1}}{V_{GS2} - V_{GS1}}.$$

Substituting the measured values,

$$g_m = \frac{15.8 - 4.401}{2.1 - 1.9}$$

$$g_m = \frac{11.399}{0.2}$$

$$g_m = 56.995 \text{ mA/V}.$$

Since

$$1 \text{ mA/V} = 1 \text{ mS},$$

we obtain

$$g_m \approx 57.0 \text{ mS}.$$

4.10 Comparison of the Transfer Characteristics

The measured values for $V_{DS} = 5 \text{ V}$, 7 V , and 9 V are very close to one another over the measured range.

For example, at $V_{GS} = 2.0 \text{ V}$,

$$I_D(5 \text{ V}) = 8.80 \text{ mA},$$

$$I_D(7 \text{ V}) = 8.82 \text{ mA},$$

and

$$I_D(9 \text{ V}) = 8.902 \text{ mA}.$$

The small differences between the three values are due to the finite output resistance and non-ideal behaviour of the practical MOSFET.

4.11 Transfer Characteristic Plot

The transfer characteristic is plotted with V_{GS} on the X-axis and I_D on the Y-axis. Three curves are plotted corresponding to

$$V_{DS} = 5 \text{ V}, \quad V_{DS} = 7 \text{ V}, \quad V_{DS} = 9 \text{ V}.$$

The curves show that I_D is very small below the threshold voltage and increases rapidly as V_{GS} is increased beyond V_T .

4.12 Results

The following values were obtained from the measured transfer characteristics:

Table 4: Calculated parameters from the transfer characteristics

Parameter	Measured Value
Threshold voltage from onset, V_T	1.4 V
Threshold voltage from $\sqrt{I_D}$ extrapolation	1.48 V
Transconductance, g_m	57.0 mS
Operating point	$V_{GS} = 2.0 \text{ V}$, $V_{DS} = 5 \text{ V}$

5 Comparison of Pinch-off Points

The theoretical saturation voltage is given by

$$V_{DS(\text{sat})} = V_{GS} - V_T$$

Using the experimentally determined threshold voltage $V_T = 1.4 \text{ V}$, the theoretical and measured pinch-off voltages are compared below.

Table 5: Comparison of experimental and theoretical pinch-off voltages

V_{GS} (V)	Experimental $V_{DS(\text{sat})}$ (V)	Theoretical $V_{GS} - V_T$ (V)
2.0	0.6	$2.0 - 1.4 = 0.6$
1.8	0.4	$1.8 - 1.4 = 0.4$
1.6	0.2	$1.6 - 1.4 = 0.2$

The experimental pinch-off voltages are in close agreement with the theoretical values obtained from $V_{GS} - V_T$.

6 Determination of Amplification Factor

The amplification factor is given by

$$\mu = g_m r_d$$

At the operating point $V_{GS} = 2.0 \text{ V}$,

$$g_m = 57.0 \text{ mS}$$

and from the output characteristics,

$$r_d = 13.33 \text{ k}\Omega$$

Therefore,

$$\begin{aligned}
\mu &= (57.0 \times 10^{-3})(13.33 \times 10^3) \\
&= \mathbf{759.81} \\
&\approx \mathbf{760}
\end{aligned}$$

Hence, the amplification factor of the MOSFET is approximately

$$\mu \approx \mathbf{760}$$

7 Precautions

- Ensure that the applied V_{GS} and V_{DS} values remain within the safe operating limits of the 2N7000 MOSFET.
- Verify the circuit connections and polarity of the power supply before taking measurements.
- Avoid excessive drain current and do not exceed the maximum power dissipation of the MOSFET.
- For calculating the amplification factor $\mu = g_m r_d$, g_m and r_d should correspond to the same operating point. The average values of g_m or r_d obtained from different operating points should not be used for calculating μ .

8 Overall Results

Table 6: Overall Experimental Results

Parameter	Measured / Calculated Value
Threshold voltage, V_T (onset method)	1.40 V
Threshold voltage, V_T ($\sqrt{I_D}$ extrapolation)	1.48 V
Transconductance, g_m	57.0 mS
Drain resistance, r_d at $V_{GS} = 2.0$ V	13.33 k Ω
Drain resistance, r_d at $V_{GS} = 1.8$ V	181.82 k Ω
Drain resistance, r_d at $V_{GS} = 1.6$ V	1.43 M Ω
Amplification factor, μ	760

9 Conclusion

The output characteristics of the 2N7000 NMOSFET demonstrated clear triode and saturation regimes, with drain current stabilizing beyond pinch-off. The experimental pinch-off points aligned closely with the theoretical boundary $V_{DS(\text{sat})} = V_{GS} - V_T$. From

the saturation slope, the drain resistance r_d was evaluated, showing an inverse dependence on the applied gate-source bias.

From the transfer characteristics, the threshold voltage was determined to be $V_T \approx 1.4\text{--}1.48$ V via turn-on onset and $\sqrt{I_D}$ extrapolation. At the operating point of $V_{GS} = 2.0$ V and $V_{DS} = 5$ V, the transconductance was computed as $g_m \approx 57.0$ mS. Combining these parameters yielded an amplification factor of $\mu \approx 760$, confirming standard mode operation.